

Markus R. Mosbech

Problem solving | Coding | Communication

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🔗 [MarkMos](#)

2025-2026

Postdoctoral Researcher, TTP @ KIT, Karlsruhe

Fundamental Research | Student Supervision | Statistical Inference

Research into the fundamental particles of our universe, including mathematical modeling, code use and implementation, and writing research papers, as well as supervising students at the bachelor and master level.

2023-2025

Postdoctoral Researcher, TTK @ RWTH, Aachen

Code Development | Technical Writing | High Performance Computing

Similar responsibilities as at KIT, as well as teaching at the bachelor and master level, and presenting my work at international conferences.

2020-2023

PhD Student, The University of Sydney, Australia

Collaboration | Project Management | Science Outreach

In addition to the research training preparing me for my later work, I served as PhD student representative on the research committee of the Dark Matter Centre of Excellence, as well as its Equity, Diversity and Inclusion committee. I was also on the physics department Early Career Reseracher committee, and part of the organising committees of multiple Early Career workshops, as well as doing outreach work. For my efforts, I received the Collaboration and Centre Values award.

General Skills

- Problem solving
- Mathematical modeling
- Statistical inference
- Research
- Linux + HPC
- Git
- Data visualization
- Technical writing
- Layout + Graphic Design
 - Affinity Suite
 - Canva
 - MS Office
 - LaTeX

Programming

- Python ★★★★★
- C ★★★★★
- html ★★★★★
- bash ★★★★★

Languages

- English ★★★★★
- Danish ★★★★★
- German ★★★★★
- French ★★★★★
- Norwegian ★★★★★
- Swedish ★★★★★

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2021-2023

Teaching assistant, The University of Sydney

Astronomy modules, 'Earth to Exoplanets', 'Stars to Black Holes', and 'Big bang to darkness'. Targeting 1st year physics students, and 1st to 3rd year students outside physics.

2017-2019

MSc. Physics, Aarhus University, Denmark

Mentor for first-year students, coordinator for the student representation, student paper editorial team

2014-2017

BSc. Physics, Aarhus University, Denmark

Teaching assistant: Philosophy and Ethics of Science: Physics, Chemistry and Geology, student paper editorial team

About Me

As a trained theoretical physicist, I am skilled in **solving quantitative problems**. I can construct models, identify simplifying approximations and confirm that they remain accurate, and implement efficient numerical treatments. I can compare model predictions with data, interpret the results, and refine models if necessary.

Working with theoretical models, I have a great deal of experience with **learning and understanding new software implementations**, and teaching myself to use them.

Furthermore, I am skilled at **presenting results** both visually and in text, constructing figures, writing papers, and preparing presentations.

Outside of quantitative analysis, I am also passionate about **graphic design**, and finding ways to communicate that are both informative and aesthetically appealing.

For examples of my work, including presentations and a list of my academic publications, check out my website linked at the top of this document. For my coding work, check out my GitHub page, also linked.